(Please write your Enrolment No. immediately) Enrolment No.

MID TERM EXAMINATION

B.TECH PROGRAMMES (UNDER THE AEGIS OF USICT)

2nd Semester, March 2025

Paper Code: ES-102 Subject: Programming in C

Time: 1½Hrs. Max. Marks: 30

Note: Attempt Q.No.1 which is compulsory and any two more questions from remaining.

		Max. Marks	CO(s)
Q.No1	a) **Write an algorithm and draw flowchart to check whether a number is odd number or even number.**	2	CO1
	b) **How does a pre-processor work?**	2	CO1

	c) **Predict the output of the following**
	# include<stdio.h>
	void main( )
	{
	int a=4,b,c;
	b = --a;
	c = a--;
	printf("%d %d %d",a,b,c);
	return 0;
	}
		2	CO2

	d). What are basic data types supported in the C Programming Language? **e)** What is the difference between the local variable and global variable in C?	2 2	CO3
2	a) **Predict the output of the following** **# include<stdio.h>** **void main( )** { int i = 1, j = 1 ; for (; ;) { if (i > 5) break ; else j += i ; printf ("%d\n", j) ; i += j ; return 0 ; } }	5 5	CO2
	b) **Write a c Program to print Factorial of a numbers without recursion.**		
3	a) **Define an array. Write a C program to print addition of two matrices of order 3.**	5	CO2
	b) **How do u use a break and goto statement in C.**	5	CO3
4	a) **Write a program to *calculate* transpose of matrices.** b) **Write 2 different C functions to compute area and perimeter of a triangle whose sides a, b, and c are given by user as inputs.**	5 5	CO2 CO3

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Time: 1½Hrs. Max. Marks: 30

Note: Attempt Q.No.1 which is compulsory and any two more questions from remaining.

		Max. Marks	CO(s)
Q.No1	a) What is an algorithm? Write an algorithm to calculate factorial of a number.	2	CO1
	b) Explain keywords & identifiers with example.	2	CO1
	c) Predict the output of the following #include<stdio.h> int main() { int i = 2, j = 3, k, l ; float a, b ; k = i / j * j ; l = j / i * i ; a = i / j * j ; b = j / i * i ; printf ("%d %d %f %f\n", k, l, a, b) ; return 0;}	2	CO2
	d) Why do we need a break and goto statement. Justify. e)What is the difference between the local variable and global variable in C.	2 2	CO3
2	a) What is an operator. Explain different types of operators used in c programming. b) Any year is input through the keyboard. Write a program to determine whether the year is a leap year or not.	5 5	CO2
3	a) Define an array. Write a Program to print the Maximum and Minimum element in an array. b) Write two different C functions to compute area and perimeter of a triangle whose sides a, b, and c are given by user as inputs. Formula to compute perimeter = a + b + c Formula to compute area = [s(s-a)(s-b)(s-c)]0.5 Where s = 0.5 * (a+b+c)	5 5	CO2 CO3
4	a) Write a program to check whether a string is a palindrome or not. b) Write a C program to check if a number has three consecutive 5s. If yes, print YES, else print NO.	5 5	CO2 CO3

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Note: Attempt Q.No.1 which is compulsory and any two more questions from remaining.

		Max. Marks	CO(s)
Q.No1	a) **Write an algorithm and draw flowchart to check whether a number is odd number or even number.**	2	CO1
	b) **How does a pre-processor work?**	2	CO1

	c) **Predict the output of the following**
	# include<stdio.h>
	void main( )
	{
	int a=4,b,c;
	b = --a;
	c = a--;
	printf("%d %d %d",a,b,c);
	return 0;
	}
		2	CO2

	d). What are basic data types supported in the C Programming Language? **e)** What is the difference between the local variable and global variable in C?	2 2	CO3
2	a) **Predict the output of the following** **# include<stdio.h>** **void main( )** { int i = 1, j = 1 ; for (; ;) { if (i > 5) break ; else j += i ; printf ("%d\n", j) ; i += j ; return 0 ; } }	5 5	CO2
	b) **Write a c Program to print Factorial of a numbers without recursion.**		
3	a) **Define an array. Write a C program to print addition of two matrices of order 3.**	5	CO2
	b) **How do u use a break and goto statement in C.**	5	CO3
4	a) **Write a program to *calculate* transpose of matrices.** b) **Write 2 different C functions to compute area and perimeter of a triangle whose sides a, b, and c are given by user as inputs.**	5 5	CO2 CO3

(Please write your Exam Roll No.) *Exam Roll No.....*

END TERM EXAMINATION

SECOND SEMESTER [B.TECH] JUNE 2024

Paper Code: ES-102 ***Subject- Programming in 'C'***

Time: 3hours **Max. Marks: 60**

Note: Attempt five questions in all including Q.No.1 which is compulsory. Select one question from each unit.

Q1 Attempt **all** questions: **[5x4=20]**

a) Difference between while and do-while loop?
b) Define different types of sorting?
c) Define pre-processor and its usage in programming?
d) Explain the basic structure and compilation process of C program?
e) Difference between structure and union?

UNIT-I

Q2 a) What is a token? Explain the different types of tokens like keyword, identifiers, constants, strings with examples? **[5]**

b) Difference between flowchart and algorithm? Write an algorithm and flowchart to find the factorial of a number? **[5]**

Q3 a) Define Operator? Explain with example different types of operator arithmetic, logical, bitwise and ternary operator? **[6]**

b) Find the output of the following program: - **[4]**

```
Int main()
{Char srt1[20]= "Hello", Str2[20]= "World";
Printf("%s\n", strcpy(str2,strcat(str1,str2)));
Printf("s\n", strrev(str1));
Printf("s\n", strlen(str2));
return 0;}
```

UNIT-II

Q4 a) Define Recursion function? Write a C program to find the Fibonacci series using recursion? **[5]**

b) Define an array? Write the advantages of an array? Write a C program to transpose of amatrix? **[5]**

Q5 a) Explain call by value and call by reference with example? **[5]**

b) Define string? How it can be declared and initialized? Explain any five-string function with example? **[5]**

UNIT-III

Q6 a) Explain File Handling? What is the need for file handling? Discuss file access modes for text files: r, a, w, r+, a+, w+? **[5]**

b) What is the need of a structure? How will you access the members of a structure? Write a program to read and print the details of a person using Structure? **[5]**

Q7 a) Define Pointer with an example? How to access a variable through pointer? Write a C program to add two variables using Pointer? **[5]**

b) Define File Handling and explain its operations? **[5]**

P.T.O.

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MID TERM EXAMINATION

B.TECH PROGRAMMES (UNDER THE AEGIS OF USICT)

2nd Semester, April 2024

Paper Code: ES-102 **Subject: Programming in C**

Time: 1½ Hrs. **Max. Marks: 30**

Note: Attempt Q. No. 1 which is compulsory and any two more questions from remaining.

Q. No.	Question	Max. Marks	CO(s)
1 (a)	Describe the compilation process in C programming language.	2	1
(b)	What is variable? Discuss the visiblity, scope, and initial value of a variable.	2	1

(c)	Find the output for the following C code
	#include <stdio.h>
	int main() {
	int i;
	for(i=2; i<5;i++)
	{
	switch(i)
	{
	case 2: printf("A\n");
	continue;
	case 4:
	printf("B\n");
	break;
	default:
	printf("C\n");
	}
	}
	return 0;
	}
		2	2

(d)	Write difference between entry-controlled loop and exit controlled loop.	2	2
(e)	What is string? Write a program to find the length of a string?	2	3
2 (a)	What is ternary operator? Design a flowchart and write a C program using ternary operator to check whether a given number is odd or even.	5	1

(b)	What is an operator? Describe various operators used in C language? Find the output of the following C program code.
	#include <stdio.h>
	int main() {
	int x = -2%10+17/3+10;
	printf("%o", x);
	return 0;
	}
		5	1